

Question		Solution	Marks
5	(a)	$n = 110 / 0.032$ or $110000 / 32$ or 3440 $pV = nRT$ $T = (1.0 \times 10^5 \times 85) / (8.31 \times (110 / 0.032)) = 300 \text{ K}$	C1 C1 A1
	(b)	$E = mc\Delta\theta$ $= 110 \times 0.66 \times 50$ $= \mathbf{3600 \text{ J}}$	C1 A1
	(c)	Any 3 from: • molecule collides with wall • momentum of molecule changes during collision (with wall) • force on molecule so force on wall • many forces act over surface area of container exerting a pressure	B3
	(d)	$KE \propto T$ $v \propto \sqrt{T}$ $ratio = \sqrt{(350 / 300)}$ $= \mathbf{1.1}$	C1 A1

Question			Solution	Marks
6	(a)	any two points from:		B2